Encapsulation

```
package com.cait.employee;

class Employee {

   // Private Data Members
   private int id;
   private String name;
   private int age;

   //  employee2.setId(102);
   public void setId(int id) {
         //employee2.id = 102;
      this.id = id;
   }

   public void setName(String name) {
      this.name = name;
   }

   public void setAge(int age)
   {
         if(age>=18)
         {
          this.age = age;
         }

   }

   // Getter Methods
   public int getId() {
      return id;
   }

   public String getName() {
      return name;
   }
```

```
    public int getAge() {
        return age;
    }

    // Display Method
    public void display() {
        System.out.println("Employee Details");
        System.out.println("----------------");
        System.out.println("Id    : " + getId());
        System.out.println("Name  : " + getName());
        System.out.println("Age   : " + getAge());
    }
}



class EmployeeMain {

    public static void main(String[] args) {

        // Employee 1
        Employee employee1 = new Employee();
        employee1.setId(101);
        employee1.setName("Priya");
        employee1.setAge(25);

        // Employee 2
        Employee employee2 = new Employee();
        employee2.setId(102);
        employee2.setName("Rahul");
        employee2.setAge(28);

        // Employee 3
        Employee employee3 = new Employee();
        employee3.setId(103);
        employee3.setName("Anitha");
```

```
        employee3.setAge(30);

        // Employee 4
        Employee employee4 = new Employee();
        employee4.setId(104);
        employee4.setName("Karthik");
        employee4.setAge(27);

        // Employee 5
        Employee employee5 = new Employee();
        employee5.setId(105);
        employee5.setName("Meena");
        employee5.setAge(26);

        // Display Employee Details
        employee1.display();
        System.out.println();

        employee2.display();
        System.out.println();

        employee3.display();
        System.out.println();

        employee4.display();
        System.out.println();

        employee5.display();

        employee1.getAge();

    }
}
```

Pen Example

```
class Pen {

   public String color;
   public String type;
   public String brand;

   void display() {
      System.out.println("Brand : " + brand);
      System.out.println("Color : " + color);
      System.out.println("Type  : " + type);
      System.out.println("-------------------------");
   }
}

class PenMain {

   public static void main(String[] args) {

      // Object 1
      Pen pen1 = new Pen();
      pen1.color = "Blue";
      pen1.type = "Ball Pen";
      pen1.brand = "Cello";

      // Object 2
      Pen pen2 = new Pen();
      pen2.color = "Black";
      pen2.type = "Gel Pen";
      pen2.brand = "Reynolds";

      // Object 3
      Pen pen3 = new Pen();
      pen3.color = "Red";
      pen3.type = "Ink Pen";
      pen3.brand = "Parker";
```

```
        // Object 4
        Pen pen4 = new Pen();
        pen4.color = "Green";
        pen4.type = "Marker";
        pen4.brand = "Camlin";

        // Object 5
        Pen pen5 = new Pen();
        pen5.color = "Purple";
        pen5.type = "Sketch Pen";
        pen5.brand = "Faber-Castell";

        pen1.display();
        pen2.display();
        pen3.display();
        pen4.display();
        pen5.display();
        System.out.println(pen1.brand);
    }
}
```

Student Example

```
class Student {

    public int id;
    public String name;
    public int age;

    public void display() {

        System.out.println("Student Details");
        System.out.println("----------------");
        System.out.println("Id   : " + id);
        System.out.println("Name : " + name);
        System.out.println("Age  : " + age);
```

```
        System.out.println();
    }
}

public class StudentMain {

    public static void main(String[] args) {

        Student student1 = new Student();
        student1.id = 101;
        student1.name = "Abi";
        student1.age = 23;

        Student student2 = new Student();
        student2.id = 102;
        student2.name = "John";
        student2.age = 24;

        Student student3 = new Student();
        student3.id = 103;
        student3.name = "Sachin";
        student3.age = 25;

        Student student4 = new Student();
        student4.id = 104;
        student4.name = "Priya";
        student4.age = 22;

        Student student5 = new Student();
        student5.id = 105;
        student5.name = "Rahul";
        student5.age = 26;

        student1.display();
        student2.display();
        student3.display();
        student4.display();
```

```
        student5.display();
    }
}
```